

clean price → price excluding accrued int

dirty price → clean price + accrued int

Yields

Zero-coupon spot yield	discount using zero-coupon spot yield → spot rate / zero rate
Bond yield	discount using single int rate to give mv → gross redemption yield
Par yield	coupon rate that makes value of bond = nominal value → prevailing pattern of spot rates

Bootstrapping: Derive a Zero coupon yield curve

- Construct from observed prices of coupon-bearing bond
- build pattern of ZC yields that is consistent with mp of conventional bonds
 - done by linear interpolation between observed values
 - find ZC yield recursively
 - ZC yield for bond with shortest outstanding duration
 - Use info to find ZC yield for next shortest bond duration etc.
- ⊙ Assumes bond market = arbitrage free

Forward Rates:



$$R_f = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1} \rightarrow \text{is continuously compounded}$$

Add + subtract $R_1 T_1$

$$R_f = R_2 + (R_1 - R_2) \frac{T_1}{(T_2 - T_1)}$$

→ As T_2 approaches T_1
Common value T

$$R_f = R + T \frac{\partial R}{\partial T}$$

→ zero rate for maturity T
→ instantaneous forward rate

CHAPTER 12: VALUATION OF INVESTMENTS

Forward Rate Agreement

$$V = L(R_K - R_f)(T_2 - T_1) e^{-R_f T_2}$$

→ value to lender

-V = value to borrower

- Assumptions
- market = arbitrage free
 - no taxes
 - no transaction costs

NOT continuously compounded

R_K = agreed int rate for between T_1 + T_2
 R_f = forward rate between T_1 + T_2

Interest Rate Swaps

$$V_{\text{swap}} = B_{3M} - B_{6M}$$

value to party lent variable + borrowed fixed

$$B_{3M} = \sum_{i=1}^n K e^{-r_i t_i} + L e^{-r_n t_n}$$

→ r_i = continuously compounded LIBOR zero rate for t_i

$$B_{6M} = (L + K^n) e^{-r_n t_n}$$

→ Floating rate payment to be made at time t_n
before payment date
after payment date = L

Swap: series of forward rate agreements

1. calculate forward rate for each LIBOR rate that determines CF
2. Calculate swap CF on assumption that LIBOR = forward rates
3. Set swap value = PV of CF

Interest Rate Futures

$$\text{Contract price} = 10,000(100 - 0.25(100 - Z))$$

→ 3-month contract

Z = quoted price on futures contract

R = actual int rate

$(100 - Z)\%$ = futures int rate convertible quarterly

→ price $Z = 100 - R$

Tick size = 1 basis point

Tick value = change in contract price

Relationship: Futures + Forward; prices

constant int rate $\rightarrow$ equal CF value (fixed)
 $\rightarrow$ equal prices

varying int rate $\rightarrow$ unequal prices

- price of underlying = $\ominus$ ively correlated with int rate

long future = more attractive than long forward

Futures price > Forward price

- price of underlying = $\ominus$ ively correlated with int rate

long forward = more attractive than long future

Futures Price < Forward Price

Short term interest rate

Futures rate > forward rate

$$\text{Forward Rate} = \text{Futures Rate} - \frac{1}{2} \sigma^2 t_1 t_2$$

convexity adjust.

- t_1 = maturity of futures contract

- t_2 = maturity of rate underlying futures contract

- σ = std dev of Δ in short-term int rate in 1 yr

Arbitrage Pricing + Hedging

Arbitrage $\begin{cases} \text{same payoff} \\ \text{same price} \end{cases}$

ELSE:

- sell expensive + buy cheaper

$\rightarrow$ unlimited return without capital expenditure

- demand for cheaper $\uparrow$

- demand for expensive $\downarrow$

Arbitrage Pricing:

no income	Income $PV = D$	Income at q pa
$F_0 = S_0 e^{rt}$	$F_0 = (S_0 - D)e^{rt}$	$F_0 = S_0 e^{(r-q)t}$

① $F_0 < S_0 e^{rt}$

- SELL asset at spot price
- INVEST proceeds at risk free rate
- enter into LONG FORWARD to buy asset at time T

$$\text{RISK FREE PROFIT} = S_0 e^{rt} - F_0$$

② $F_0 > S_0 e^{rt}$

- BORROW S_0 to buy asset
- enter into SHORT FORWARD to sell asset at time T
- REPAY loan + accumulated int

$$\text{RISK FREE PROFIT} = F_0 - S_0 e^{rt}$$

Hedging $\rightarrow$ eliminate market risk

Short hedge

Investor due to SELL
 $\therefore$ take SHORT futures position

$\downarrow$ asset price

- loss on sale
- gain on future

$\uparrow$ asset price

- gain on sale
- loss on future

Long hedge

Investor due to BUY
 $\therefore$ take LONG futures position

$\downarrow$ asset price

- loss on future
- gain on purchase

$\uparrow$ asset price

- gain on future
- loss on purchase

may not improve the overall financial outcome but does reduce risk

Basis = difference between futures price of contract used to hedge + spot price of asset to be hedged

[Futures Price - Spot Price]

RISK:

$\odot$ do not move in line due to volatility in basis

When does basis risk arise?

- asset whose price is to be hedged $\neq$ asset underlying future
 $\rightarrow$ CROSS-HEDGE
- uncertainty on date asset is to be bought/sold
- hedge requires future to be closed out well before expiration

optimal hedge ratio:

$$h = \rho \frac{\sigma_s}{\sigma_f}$$

$\rightarrow$ std dev of Δ spot price
 $\rightarrow$ std dev of Δ futures price
 $\rightarrow$ corr between ΔS and ΔF

optimal hedge ratio = 1
 $\rho = 1, \Delta S = \Delta F$

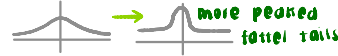
Empirical characteristics of asset prices

departure of $\begin{matrix} \text{price} \\ \text{return} \end{matrix}$ from assumptions in models

Assumptions:

1. Normality of increments in (log) asset prices:

Equity return distribution = leptokurtic



2. Δ increments in asset prices

Variance grows at slower than linear rate $\rightarrow$ long-run mean reversion

Studies show $\begin{cases} \text{non-linear dependence} \\ |\text{abs}| \text{ returns} + \text{returns}^2 = \text{autocorrelated} \end{cases}$

No serial corr evidence for prices

3. Consistency of parameters $\begin{cases} \text{drift } (\mu) \\ \text{volatility } (\sigma) \end{cases}$

returns² = non-linear dependence $\rightarrow$ heteroscedastic NOT homoscedastic

volatility varies systematically

clustered volatility $\begin{cases} \text{cluster large + small shocks} \\ \text{non-linear dependence in returns series} \end{cases}$

Relationships:

- equity values
- dividend growth rate
- inflation

Definitions:

• Forward rate	rate at which 2 parties agree to lend/borrow a specified principal over a specified future period
• Forward rate agreement	forward contract where parties agree that a specified $\bar{r}$ will apply to a specified principal during a specified future time
• Hedge	trade to reduce market risk
• Market risk	risk of losses due to unpredictable changes in the market values of assets
• Basis	difference between futures price of contract used to hedge + spot price of assets to be hedged
• Basis risk	risk that futures and spot price will not move in line due to volatility in the basis
• Optimal hedge ratio	ratio of the size of the position taken in the futures contract to the size of the exposure